Docker Deployment & Troubleshooting Guide

1. Core Concept Breakdown

Before running containers, a DevOps engineer must understand the fundamental architectural difference between an Image and a Container:

Concept	Definition	Analogy	State
Docker Image	A read-only template with instructions for creating a container.	The **blueprint** or zipped installer.	Static
Docker Container	A runnable, isolated instance of an image.	The **physical building** or active application.	Dynamic

2. Troubleshooting Permission Denied Errors

If you receive a *'permission denied while trying to connect to the Docker daemon socket'* error, your current user lacks permissions to interact with the Docker API Unix socket.

Immediate Quick Fix: Run the command using administrator privileges via sudo:

```
sudo docker run -d -p 8080:80 --name sathvik nginx
```

Permanent Fix (Run Docker without sudo): Add your Linux user account directly to the docker system group:

```
# 1. Create the docker group (if it does not exist)
sudo groupadd docker

# 2. Add your current user to the group
sudo usermod -aG docker $USER

# 3. Activate the new group configurations
newgrp docker
```

3. Complete Container Deployment Workflow

Step 3.1: Run the Container

```
docker run -d -p 8080:80 --name sathvik nginx
```

Flags: -d runs detached in background; -p 8080:80 maps host port 8080 to container port 80; --name sets container name.

Step 3.2: Verify Container Status

```
docker ps
```

Ensures the container status shows as up, running, and maps the expected active ports.

Step 3.3: Verify Application Access

```
curl http://localhost:8080
```

Validates that the web server application starts correctly and responds over the network connection.

Step 3.4: Inspect Container Logs

```
docker logs sathvik
```

Fetches application standard output and error logs to confirm successful internal initialization steps.

Step 3.5: Inspect Detailed Metadata Information

```
docker inspect sathvik
```

Returns comprehensive structural JSON metadata containing internal IP addresses, volume paths, and environment settings.